

**INSTITUTO FEDERAL DE EDUCAÇÃO, CIÊNCIA E TECNOLOGIA DE MINAS**  
**GERAIS CAMPUS SÃO JOÃO EVANGELISTA**  
**BACHAREL EM SISTEMAS DE INFORMAÇÃO**

**Disciplina: Desenvolvimento Web II**

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**Aluna: Bianca de Castro Aguiar Fontes**

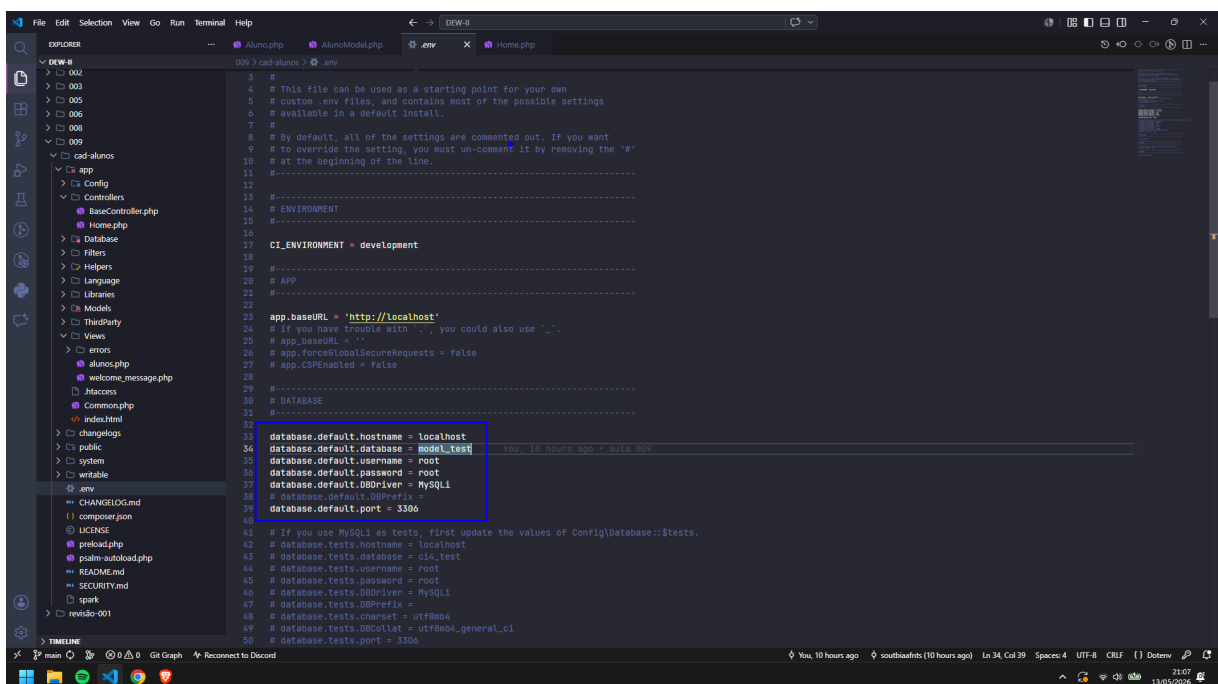
**Prática VI - Integração com Banco de dados Models**

**Objetivo**

Compreender a estrutura básica do CodeIgniter, utilizando rotas, views, models e integração com banco de dados para realizar operações de cadastro e exibição de dados. Vimos sobre Session, Flashdata, requisições, etc.

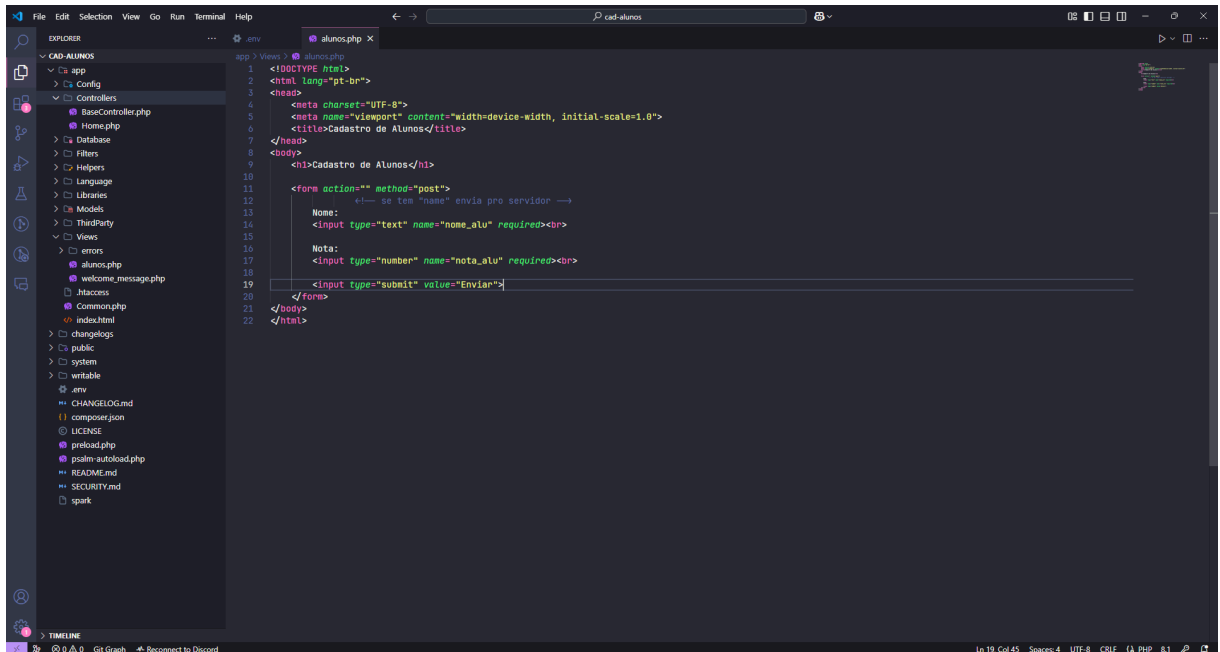
**Desenvolvimento**

3. Configuramos a configuração de conexão com banco de dados no arquivo `.env`.



```
1 #
2 # This file can be used as a starting point for your own
3 # custom .env files, and contains most of the possible settings
4 # available in a default install.
5 #
6 # By default, all of the settings are commented out. If you want
7 # to override the setting, you must un-comment it by removing the '#'
8 # at the beginning of the line.
9 #-----
10 # ENVIRONMENT
11 #-----
12 CI_ENVIRONMENT = development
13 #-----
14 # APP
15 #-----
16 app.baseURL = 'http://localhost'
17 # If you have trouble with '...', you could also use '...'
18 # app.baseURL = ''
19 # app.forceGlobalSecureRequests = false
20 # app.CSPEnabled = false
21 #-----
22 # DATABASE
23 #-----
24 database.default.hostname = localhost
25 database.default.database = model_test
26 database.default.username = root
27 database.default.password = root
28 database.default.driver = MySQLi
29 database.default.prefix =
30 database.default.port = 3306
31 #-----
32 # If you use MySQL as tests, first update the values of Config\Database::$tests.
33 # database.tests.hostname = localhost
34 # database.tests.database = ci4_test
35 # database.tests.username = root
36 # database.tests.password = root
37 # database.tests.driver = MySQLi
38 # database.tests.prefix =
39 # database.tests.charset = utf8mb4
40 # database.tests.collat = utf8mb4_general_ci
41 # database.tests.port = 3306
```

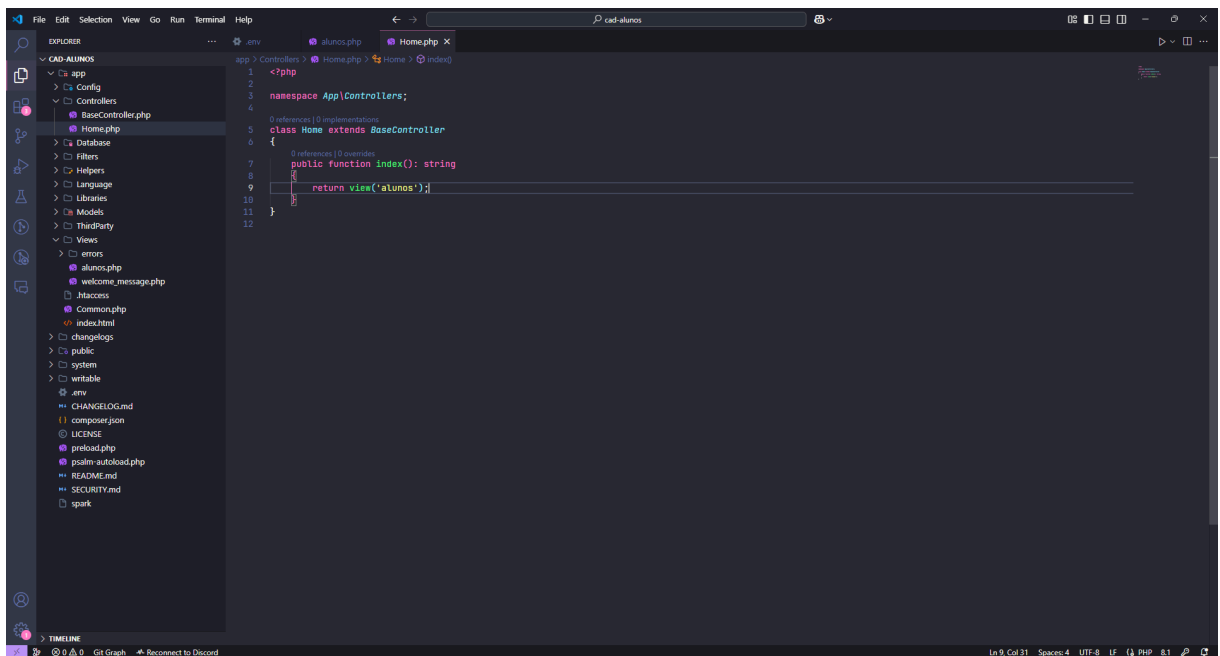
4. Criamos uma view alunos e retornamos ela no método Home::index, nessa view fizemos um formulário simples para cadastro de alunos



```

1 <?php
2 <html lang="pt-br">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Cadastro de Alunos</title>
7 </head>
8 <body>
9 <h1>Cadastro de Alunos</h1>
10
11 <form action="" method="post">
12 <!-- se tem "name" envia pro servidor -->
13 Nome:
14 <input type="text" name="nome_alu" required><br>
15
16 Nota:
17 <input type="number" name="nota_alu" required><br>
18
19 <input type="submit" value="Enviar">
20 </form>
21 </body>
22 </html>

```

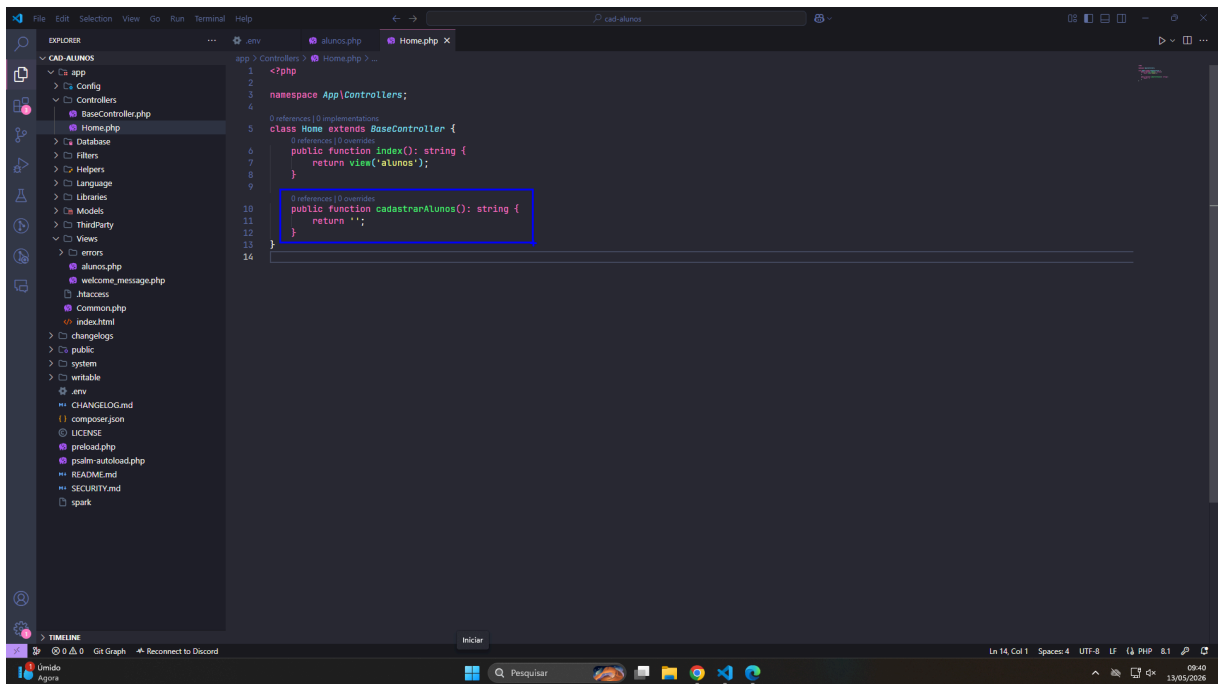


```

1 <?php
2
3 namespace App\Controllers;
4
5 class Home extends BaseController
6 {
7     // references (0 overrides)
8     public function index(): string
9     {
10         return view('alunos');
11     }
12 }

```

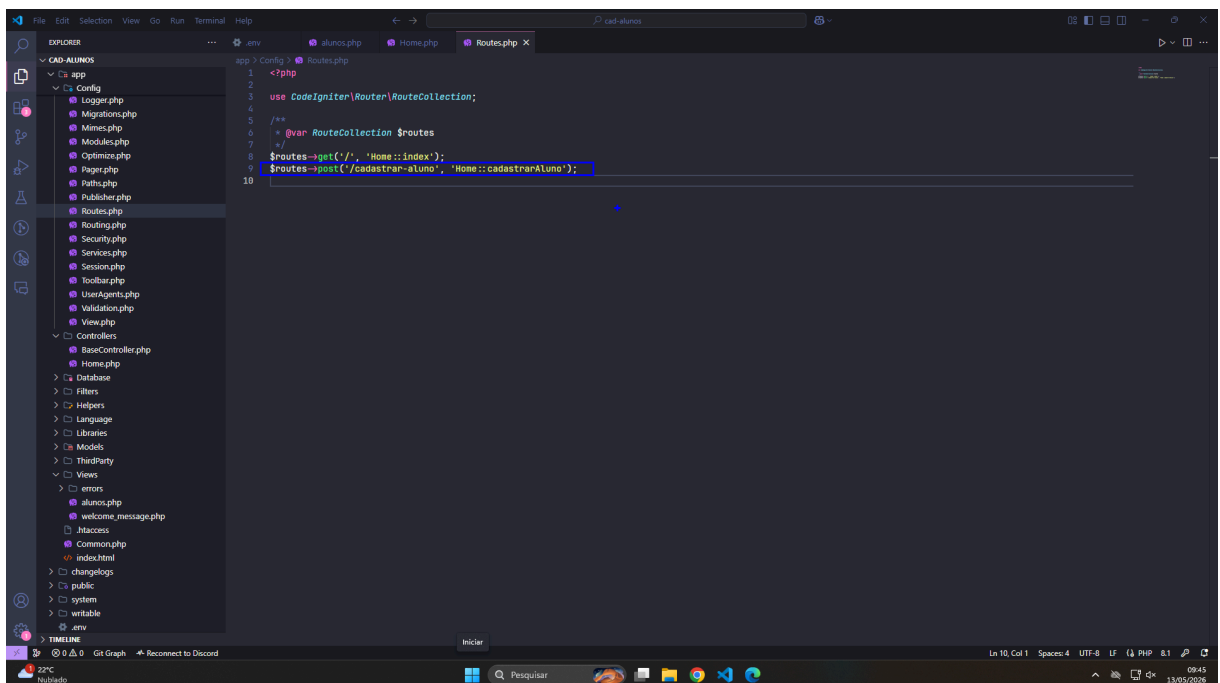
5. Criamos um outro método, o `Home::cadastrarAluno` e uma rota POST para ele usando a URI `cadastrar-aluno`.



```

1 <?php
2
3 namespace App\Controllers;
4
5 class Home extends BaseController {
6     public function index(): string {
7         return view('alunos');
8     }
9
10    public function cadastrarAlunos(): string {
11        return '';
12    }
13 }
14

```



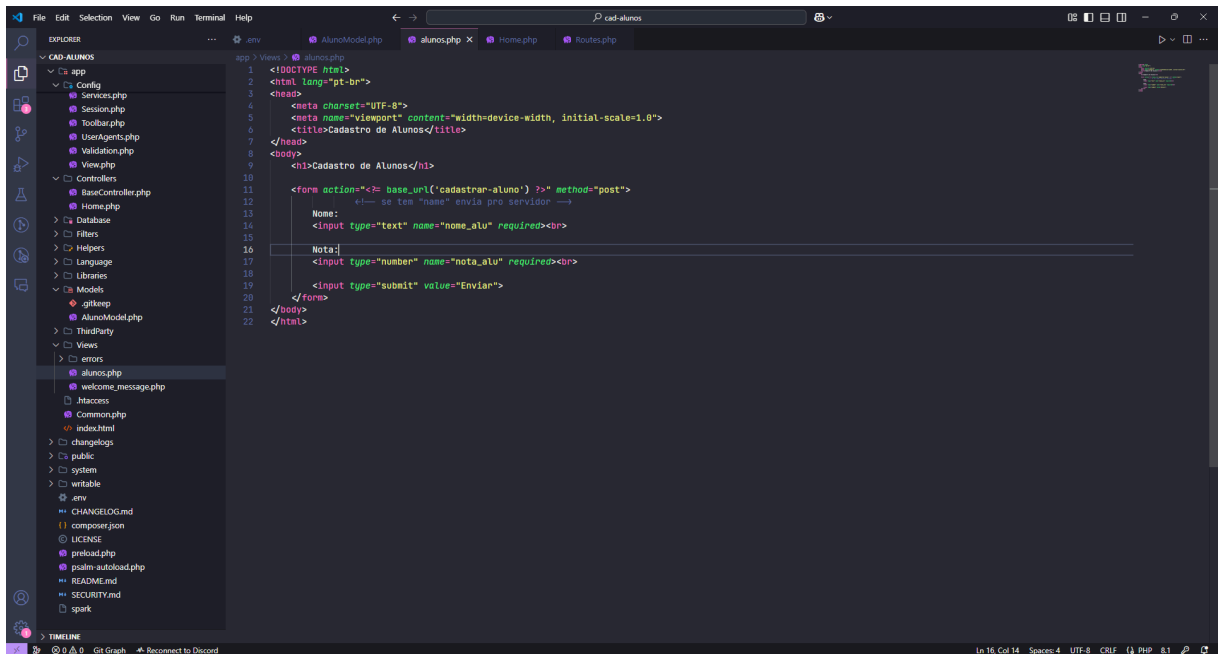
```

1 <?php
2
3 use CodeIgniter\Routing\RouteCollection;
4
5 /**
6  * @var RouteCollection $routes
7  */
8 $routes->get('/', 'Home::index');
9 $routes->post('/cadastrar-aluno', 'Home::cadastrarAluno');
10

```

Não é possível acessar a rota POST pela barra de pesquisa e não é clicável na debugbar, só é possível acessar por uma requisição.

## 6. Configuramos o atributo action do formulário na view alunos para a rota cadastrar-aluno



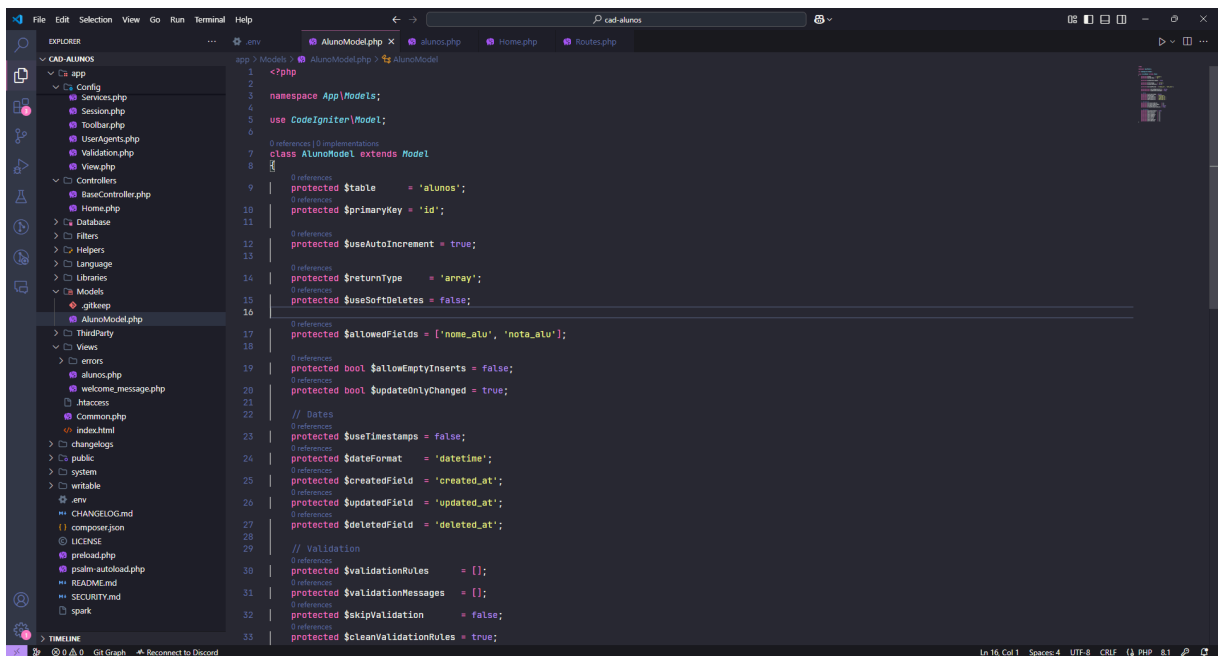
The screenshot shows the Visual Studio Code editor with the 'alunos.php' file open. The file contains an HTML form with two input fields: 'Nome' and 'Nota'. The 'Nome' field is a text input, and the 'Nota' field is a number input. The form is enclosed in a <form> tag with the following attributes: <form action=<? base\_url('cadastrar-aluno') >? method='post'>. The 'Nome' field is labeled 'Nome:' and has the attribute <input type='text' name='nome\_alu' required>. The 'Nota' field is labeled 'Nota:' and has the attribute <input type='number' name='nota\_alu' required>. The form is submitted using a button with the attribute <input type='submit' value='Enviar'>.

```

1 <?php
2 <doctype html>
3 <html lang="pt-br">
4 <head>
5 <meta charset="UTF-8">
6 <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 <title>Cadastro de Alunos</title>
8 </head>
9 <body>
10 <h1>Cadastro de Alunos</h1>
11
12 <form action=<? base_url('cadastrar-aluno') >? method="post">
13     <!-- Se tem "name" envia pro servidor -->
14     Nome:
15     <input type="text" name="nome_alu" required><br>
16
17     Nota:
18     <input type="number" name="nota_alu" required><br>
19     <input type="submit" value="Enviar">
20 </form>
21
22 </body>
23 </html>

```

## 7. Criamos o Model → AlunoModel



The screenshot shows the Visual Studio Code editor with the 'AlunoModel.php' file open. The file contains the following PHP code:

```

1 <?php
2
3 namespace App\Models;
4
5 use CodeIgniter\Model;
6
7 class AlunoModel extends Model
8 {
9     // references
10     protected $table = 'alunos';
11     // references
12     protected $primaryKey = 'id';
13     // references
14     protected $useAutoIncrement = true;
15     // references
16     protected $returnType = 'array';
17     // references
18     protected $useSoftDeletes = false;
19
20     // references
21     protected $allowedFields = ['nome_alu', 'nota_alu'];
22
23     // references
24     protected bool $allowEmptyInserts = false;
25     // references
26     protected bool $updateOnlyChanged = true;
27
28     // Dates
29     // references
30     protected $useTimestamps = false;
31     // references
32     protected $dateFormat = 'datetime';
33     // references
34     protected $createdField = 'created_at';
35     // references
36     protected $updatedField = 'updated_at';
37     // references
38     protected $deletedField = 'deleted_at';
39
40     // Validation
41     // references
42     protected $validationRules = [];
43     // references
44     protected $validationMessages = [];
45     // references
46     protected $skipValidation = false;
47     // references
48     protected $cleanValidationRules = true;
49 }

```

8. Implementamos o método `Home::cadastrarAluno` para inserir os dados no banco de dados, bem como o redirecionamento para a página inicial

```

1 <?php
2
3 namespace App\Controllers;
4
5 use App\Models\AlunoModel;
6
7 // references: 0 implementations
8 class Home extends BaseController {
9     // references: 0 overrides
10     public function index(): string {
11         return view('alunos');
12     }
13
14     // references: 0 overrides
15     public function cadastrarAluno() {
16         // recebe dados do formulário
17         $nome = $this->request->getPost('nome_alu');
18         $nota = $this->request->getPost('nota_alu');
19
20         // cria um array associativo dos atributos do BD
21         $aluno = [
22             'nome_alu' => $nome,
23             'nota_alu' => $nota
24         ];
25
26         $alunoModel = new \App\Models\AlunoModel();
27
28         // o método save, salva ou atualiza
29         if ($alunoModel->save($aluno)) {
30             return redirect()->to('/');
31         } else {
32             return redirect()->to('/');
33         }
34     }
35 }
  
```

**Cadastro de Alunos**

Nome:

Nota:

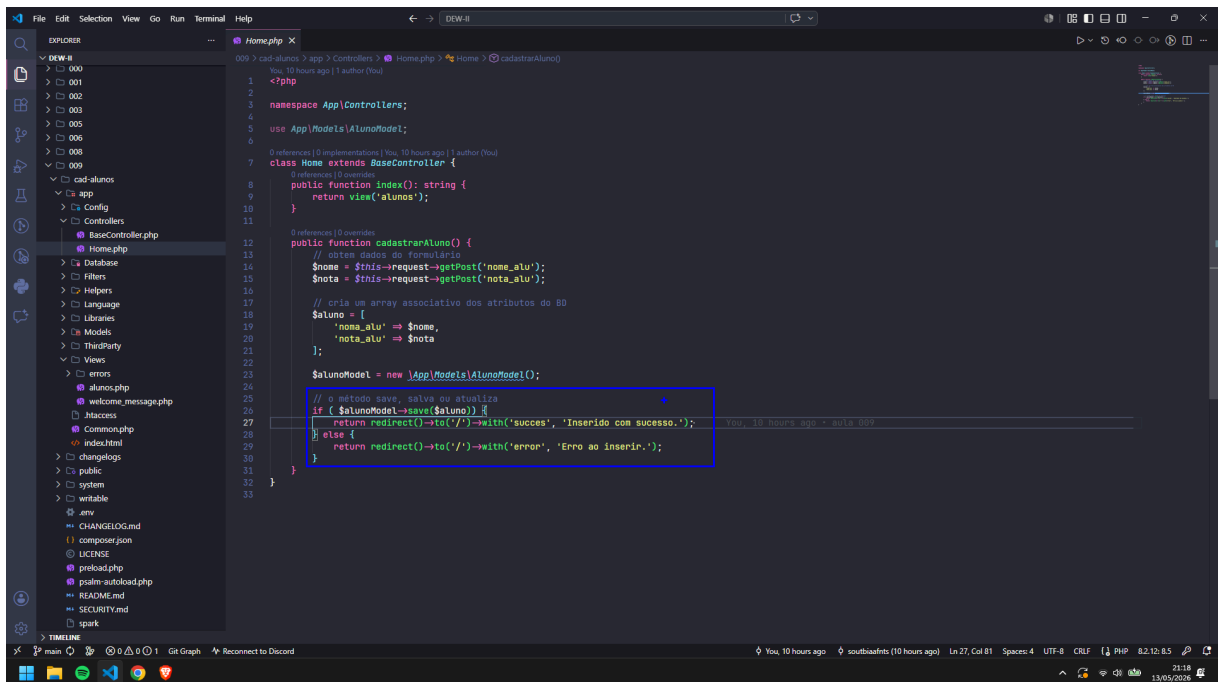
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**History**

| Action | Datetime                   | Status | Method | URL                                        | Content-Type             | Is AJAX? |
|--------|----------------------------|--------|--------|--------------------------------------------|--------------------------|----------|
| Load   | 2026-05-13 13:29:50.311346 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:29:15.190740 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:29:15.100726 | 303    | POST   | http://localhost/index.php/cadastrar-aluno | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:26:13.456223 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:22:53.070250 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:22:53.985342 | 303    | POST   | http://localhost/index.php/cadastrar-aluno | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:21:59.926457 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:11:12.059010 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:04:29.109714 | 200    | POST   | http://localhost/index.php/cadastrar-aluno | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:04:21.880320 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:04:21.759368 | 200    | POST   | http://localhost/index.php/cadastrar-aluno | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 13:03:22.137325 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:55:03.594205 | 200    | GET    | http://localhost:8080/index.php/           | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:54:29.295911 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:52:25.115592 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:51:41.647841 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:47:04.897773 | 200    | GET    | http://localhost/index.php/_hot-reload     | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:47:03.579863 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:47:03.479295 | 200    | GET    | http://localhost/index.php/_hot-reload     | text/html; charset=UTF-8 | No       |
| Load   | 2026-05-13 12:46:59.048022 | 200    | GET    | http://localhost/index.php/                | text/html; charset=UTF-8 | No       |

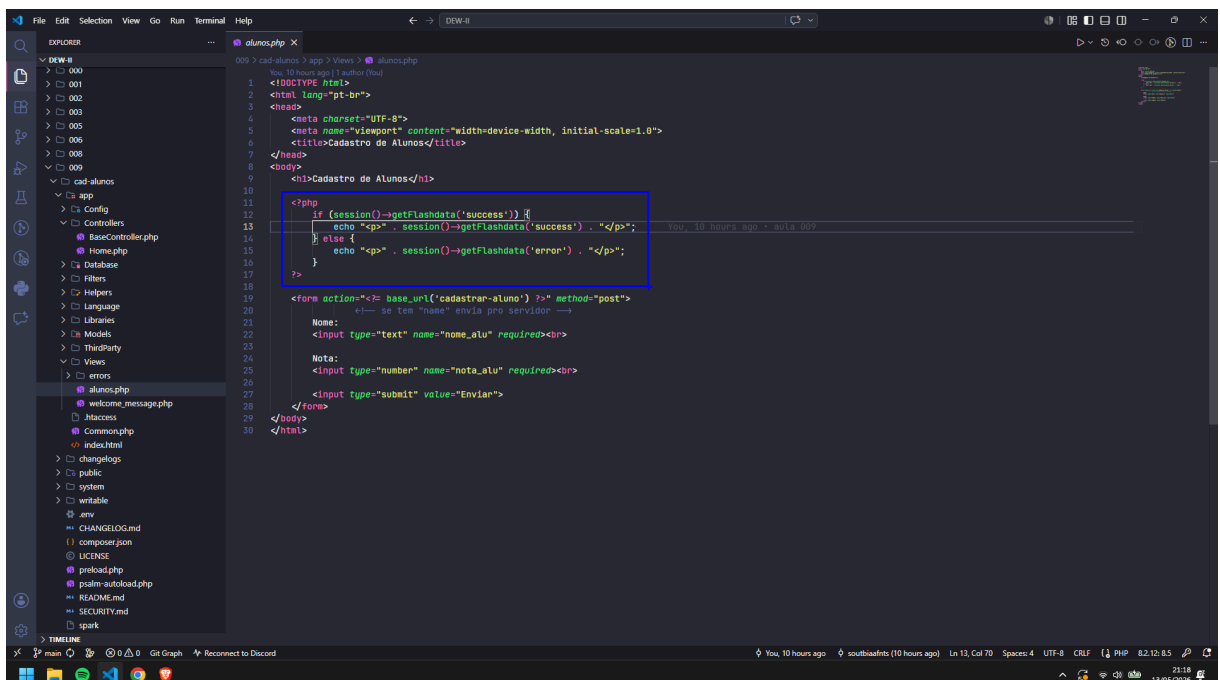
58.7 ms 5.273 MB Database Views Files 137 Routes 1 Events 2 History 20 Vars

9. Configuramos uma mensagem de sucesso e de erro e as enviamos utilizando sessões e dados flash



```

1  <?php
2
3  namespace App\Controllers;
4
5  use App\Models\AlunoModel;
6
7  class Home extends BaseController {
8
9      public function index(): string {
10         return view('alunos');
11     }
12
13     public function cadastrarAluno() {
14         // obten dados do formulário
15         $nome = $this->request->getPost('nome_alu');
16         $nota = $this->request->getPost('nota_alu');
17
18         // cria um array associativo dos atributos do BD
19         $aluno = [
20             'nome_alu' => $nome,
21             'nota_alu' => $nota
22         ];
23
24         $alunoModel = new App\Models\AlunoModel();
25
26         // o método save, salva ou atualiza
27         if ($alunoModel->save($aluno)) {
28             return redirect()->to('/')->with('sucesso', 'Inserido com sucesso.');
```



```

1  <?php
2
3  <doctype html>
4
5  <html lang="pt-br">
6
7  <head>
8      <meta charset="UTF-8">
9      <meta name="viewport" content="width=device-width, initial-scale=1.0">
10     <title>Cadastro de Alunos</title>
11 </head>
12 <body>
13     <h1>Cadastro de Alunos</h1>
14
15     <?php
16     if (session()->getFlashdata('success')) {
17         echo "<p>". session()->getFlashdata('success') . "</p>";
18     } else {
19         echo "<p>". session()->getFlashdata('error') . "</p>";
20     }
21
22     <form action="<= base_url('cadastrar-aluno') ">" method="post">
23         <!-- se tem "nome" envia pro servidor -->
24         Nome:
25         <input type="text" name="nome_alu" required><br>
26
27         Nota:
28         <input type="number" name="nota_alu" required><br>
29         <input type="submit" value="Enviar">
30     </form>
31 </body>
32 </html>
```

Não tive problemas para seguir essa aula prática também, gostei muito de entender mais sobre sessions! Infelizmente me esqueci de tirar o print da tela de resultado na aula e meu

computador não tem conseguido rodar o MySQL Workbench, mas garanto que estava tudo certo! Foi uma ótima aula.